



E-BSP

Stock assessment of Jumbo squid in the South Pacific using E-BSP

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Shanghai Ocean University

07/29/2026

> continue >



Model description

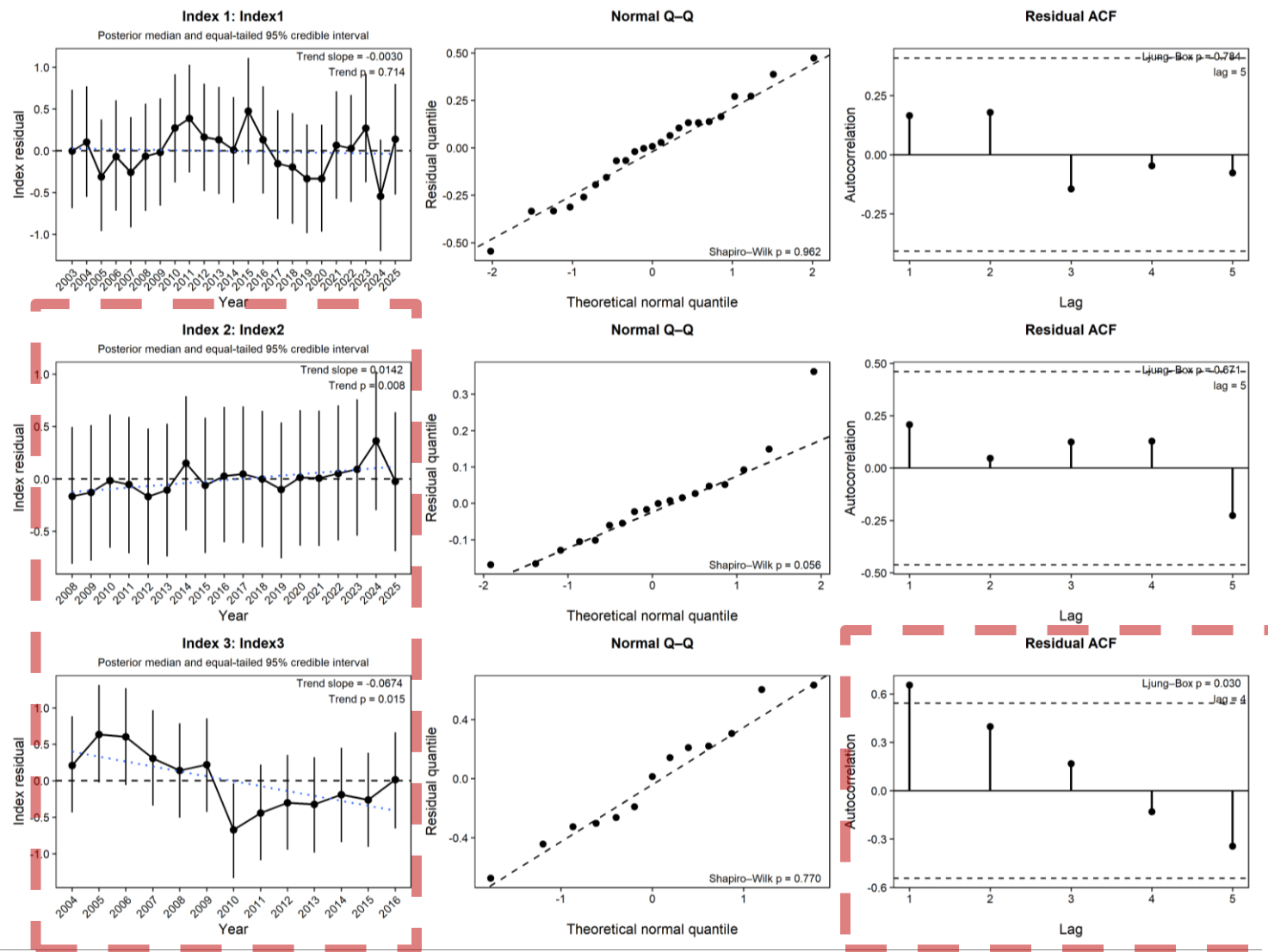
Model	Time period	CPUE series	Peruvian CPUE included	Weight assigned to Peruvian CPUE	Role in subsequent analysis
S1	2012–2025	China, Chilean artisanal, Chilean industrial	No	—	Retained; baseline for SE1–SE3
S2	2012–2025	China, Chilean artisanal, Chilean industrial, Peru	Yes	0.1	Retained; baseline for SE4–SE6
S3	2003–2025	China, Chilean artisanal, Chilean industrial	No	—	Not retained because of CPUE diagnostic issues
S4	2003–2025	China, Chilean artisanal, Chilean industrial, Peru	Yes	0.1	Not retained because of CPUE diagnostic issues

$$r_t = r \exp(\beta_r E_{r,t})$$

$$q_{i,t} = q_i \exp(\beta_{q_i} E_{q_i,t})$$

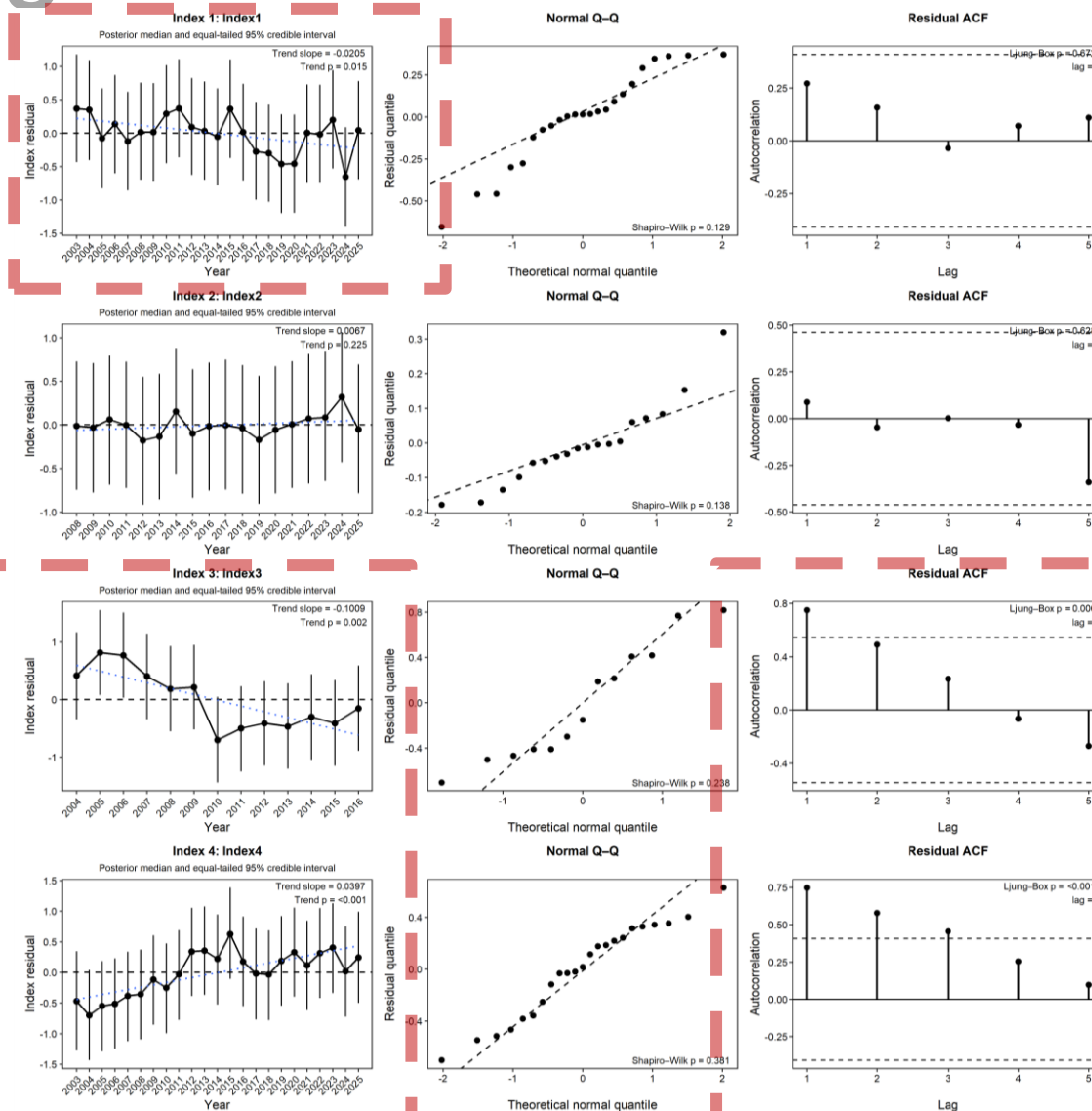
Model diagnostic

S3



Model diagnostic

S4





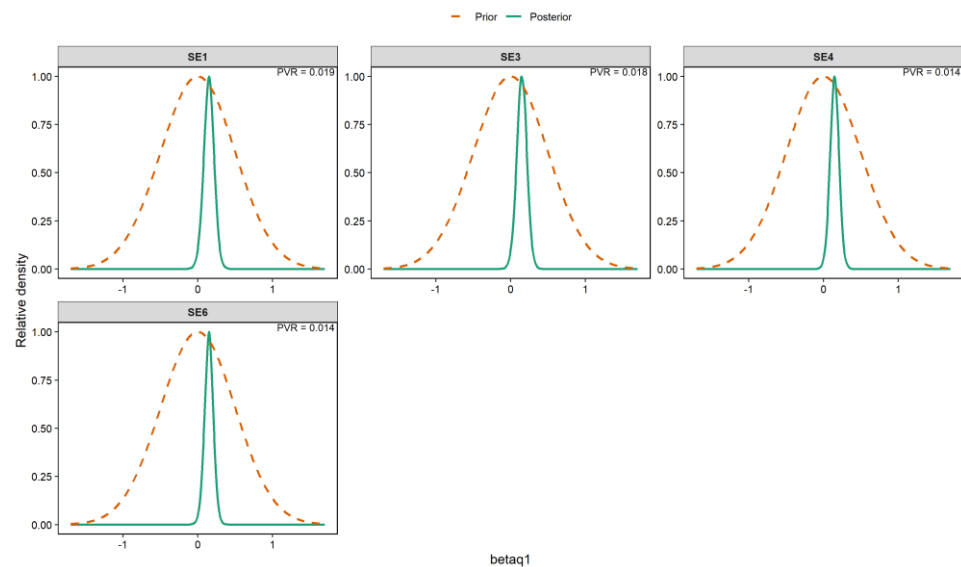
Model description

Model	Baseline model	Time period	CPUE series	Environmental effect on r	Environmental effect on q_i
SE1	S1	2012–2025	China, Chilean artisanal, Chilean industrial	Yes	Yes
SE2	S1	2012–2025	China, Chilean artisanal, Chilean industrial	Yes	No
SE3	S1	2012–2025	China, Chilean artisanal, Chilean industrial	No	Yes
SE4	S2	2012–2025	China, Chilean artisanal, Chilean industrial, Peru	Yes	Yes
SE5	S2	2012–2025	China, Chilean artisanal, Chilean industrial, Peru	Yes	No
SE6	S2	2012–2025	China, Chilean artisanal, Chilean industrial, Peru	No	Yes

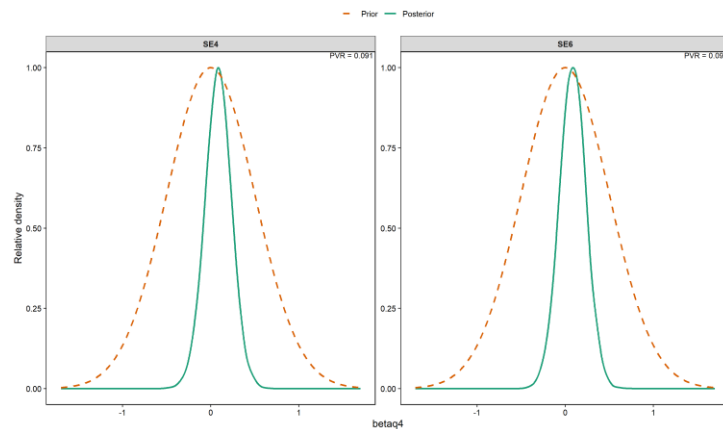


Model diagnostic

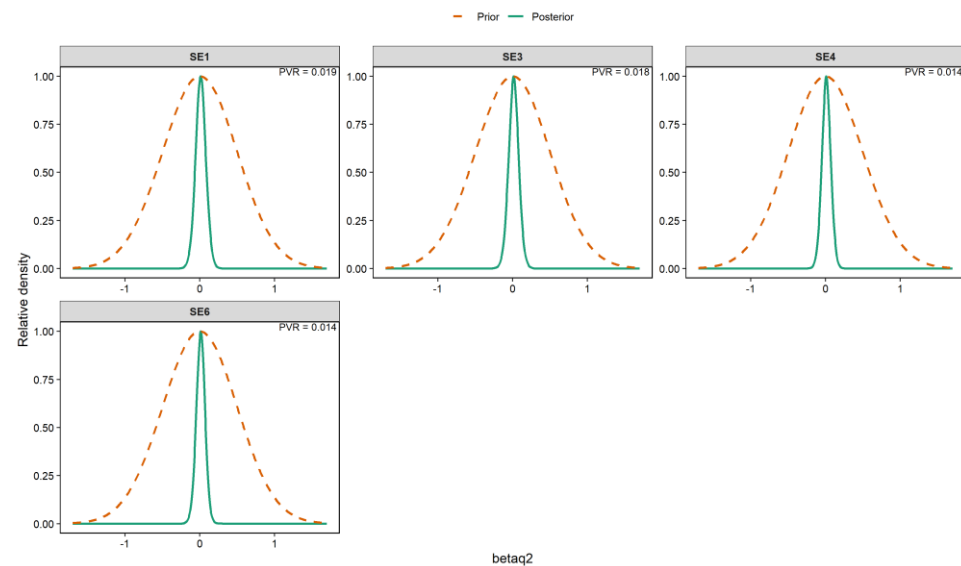
betaq1: prior and posterior across SE1-SE6



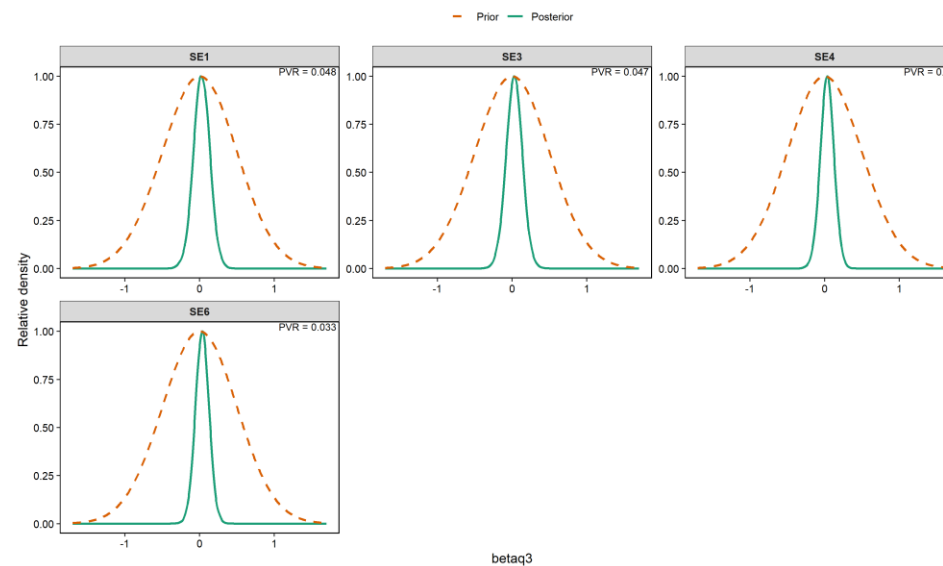
betaq4: prior and posterior across SE1-SE6



betaq2: prior and posterior across SE1-SE6

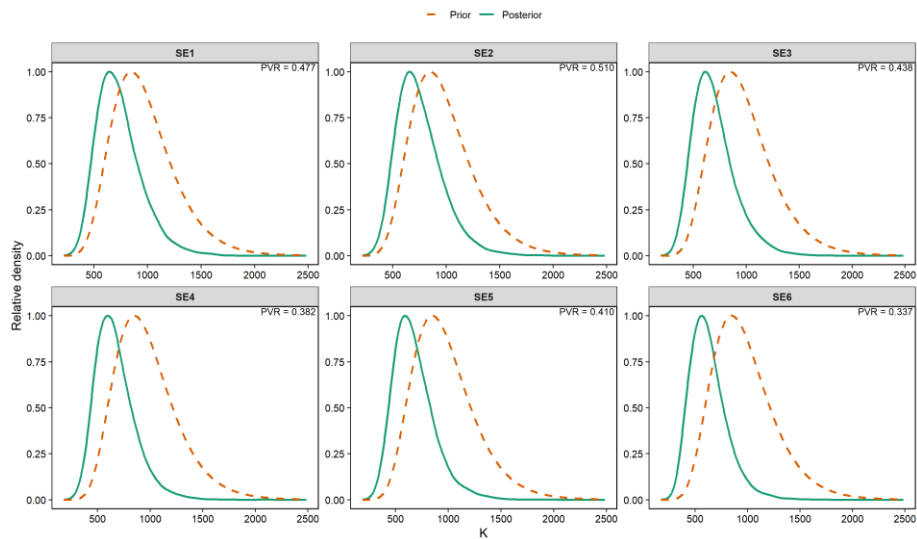


betaq3: prior and posterior across SE1-SE6

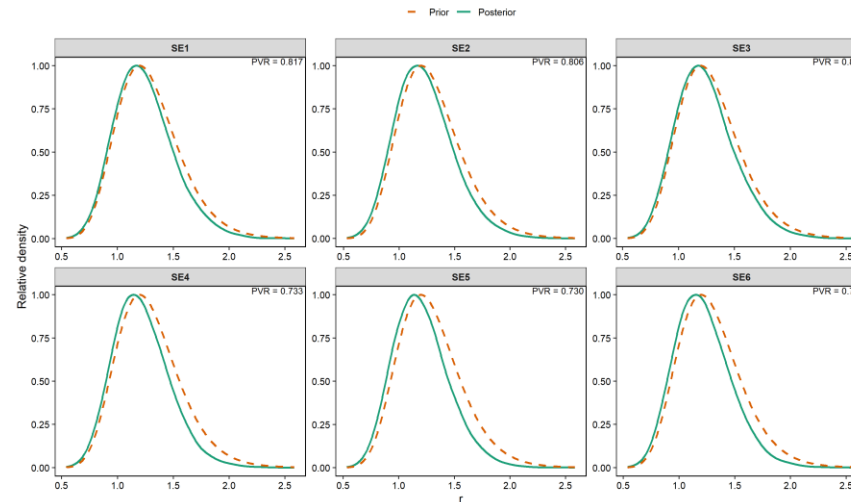


Model diagnostic

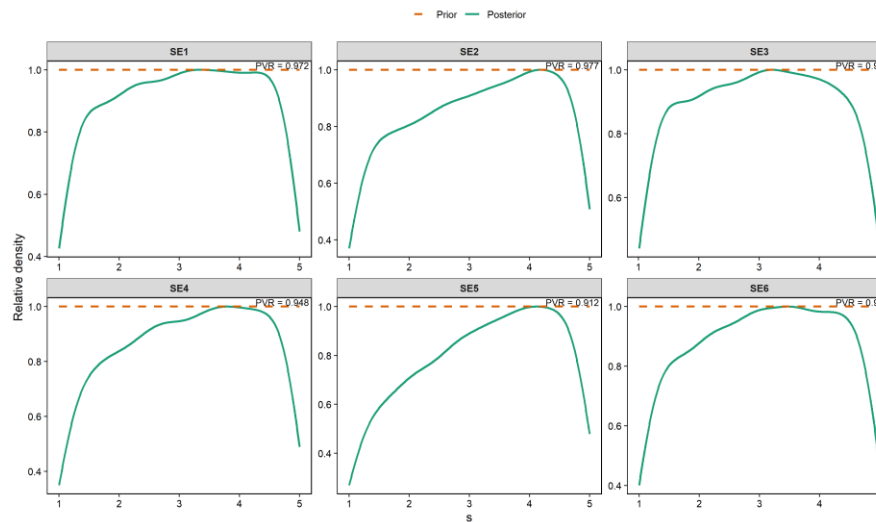
K: prior and posterior across SE1-SE6



r: prior and posterior across SE1-SE6

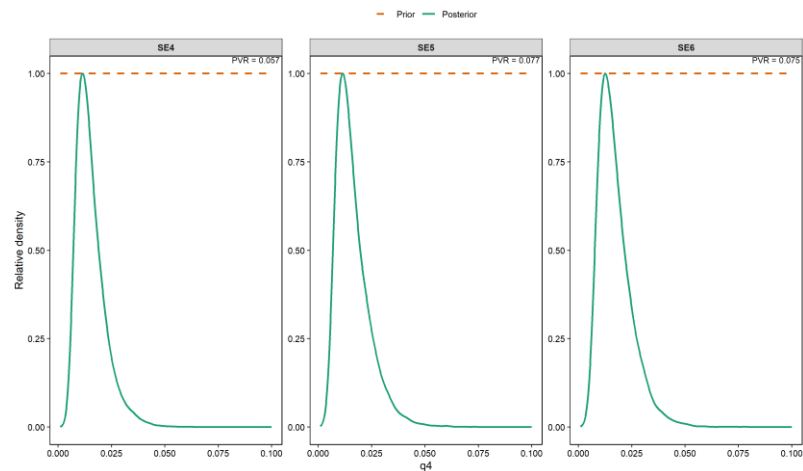


s: prior and posterior across SE1-SE6

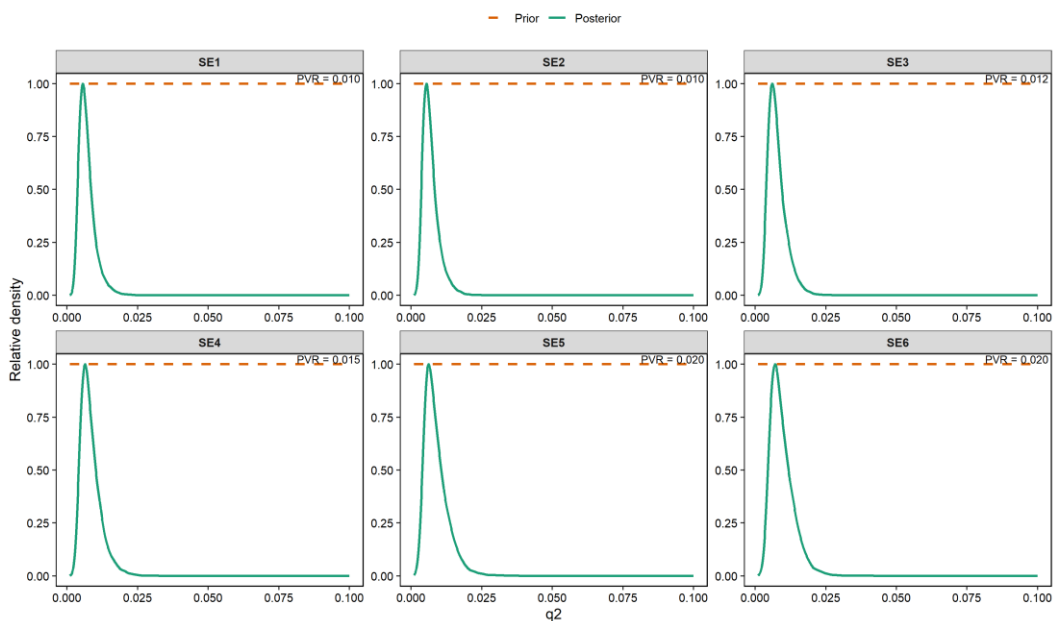


Model diagnostic

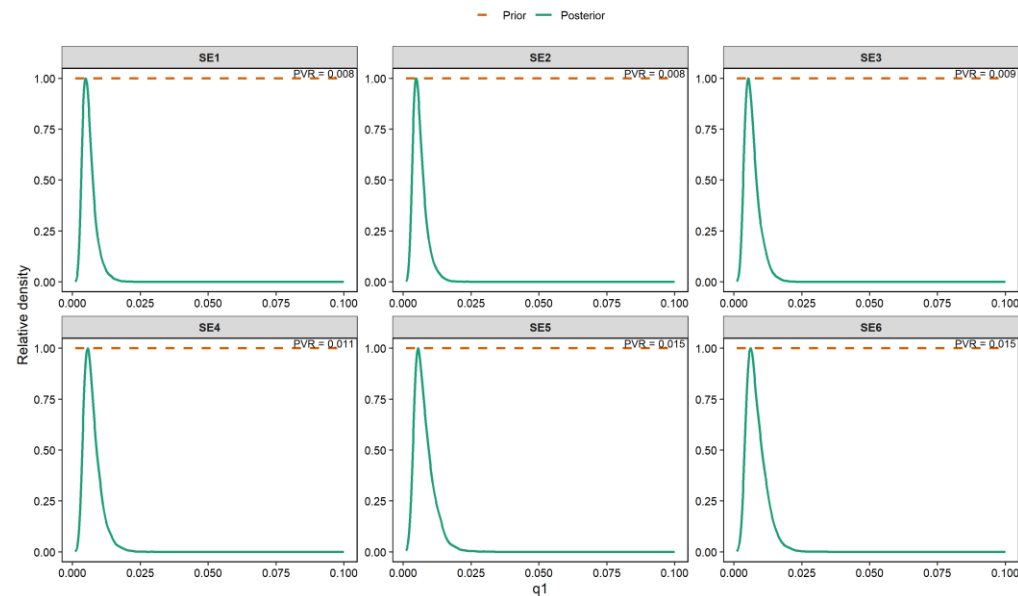
q4: prior and posterior across SE1-SE6



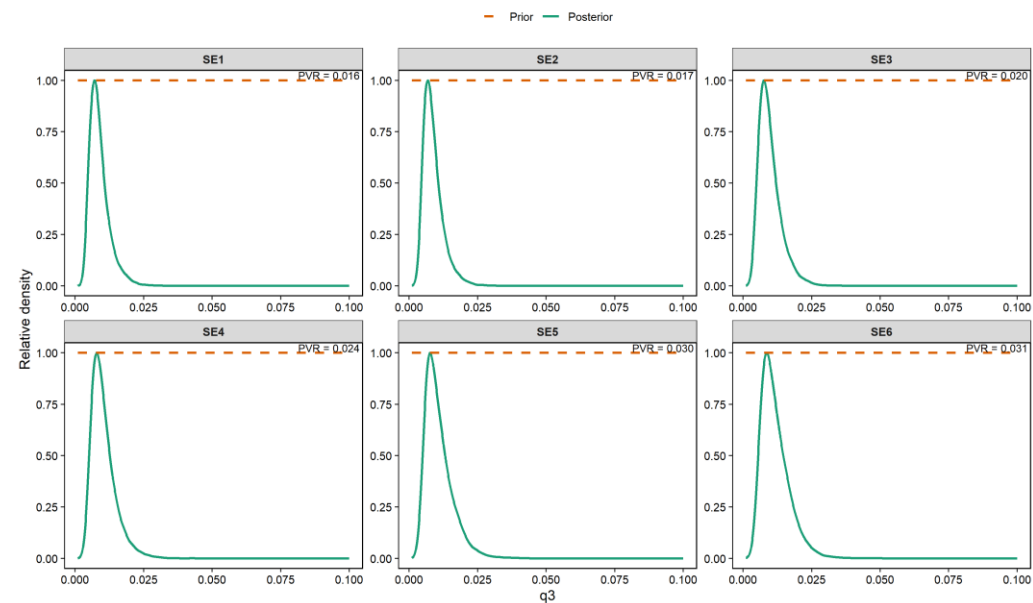
q2: prior and posterior across SE1-SE6



q1: prior and posterior across SE1-SE6

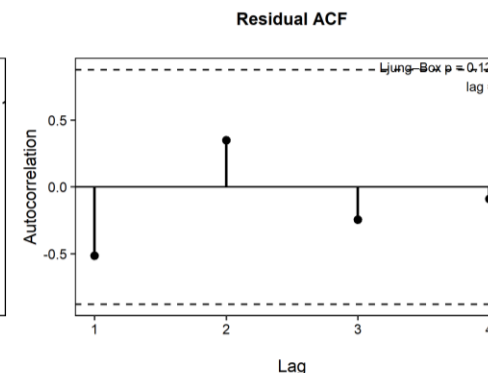
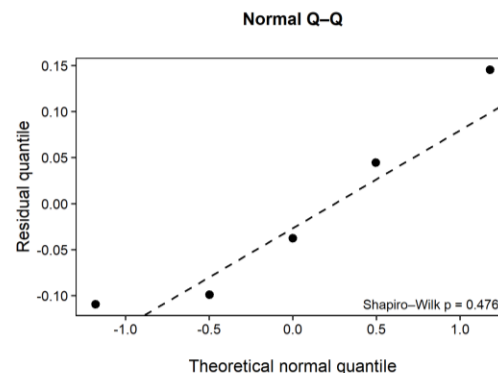
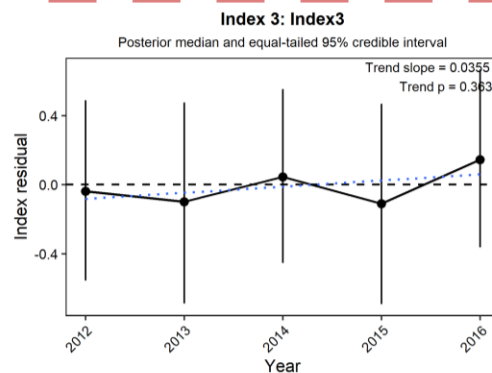
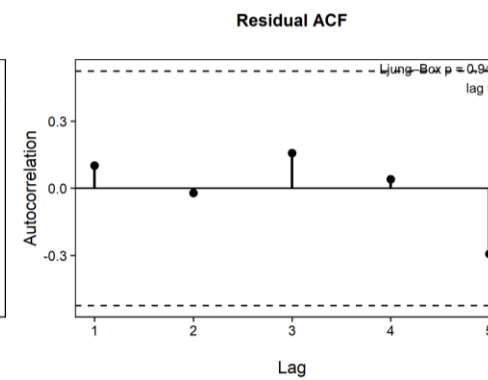
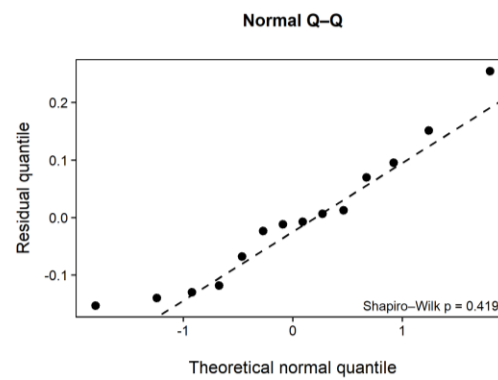
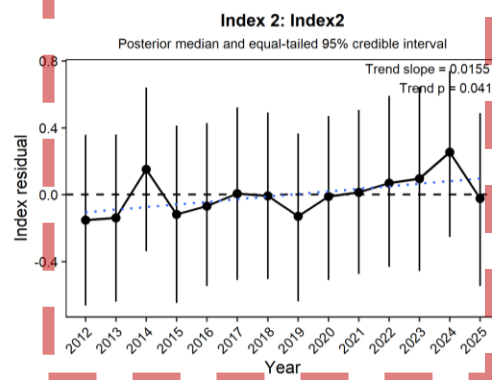
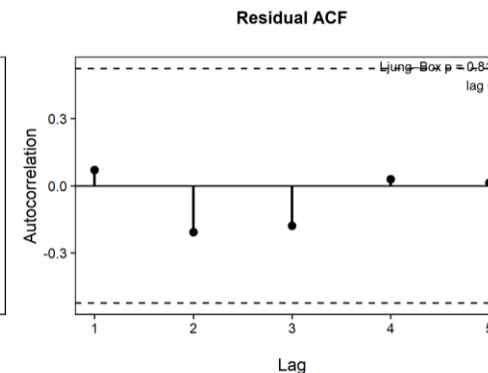
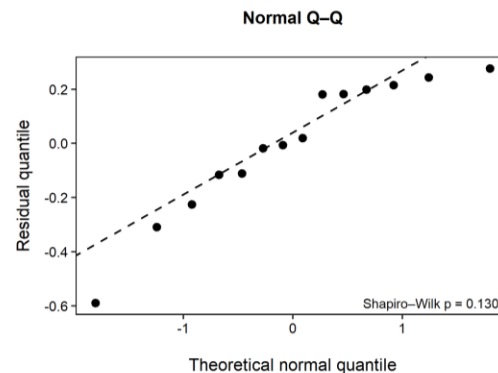
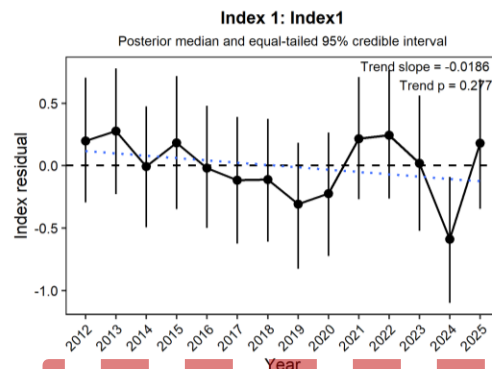


q3: prior and posterior across SE1-SE6



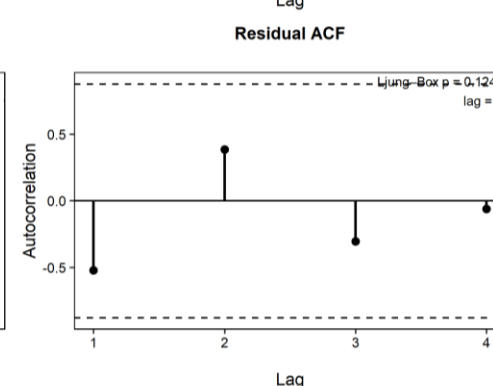
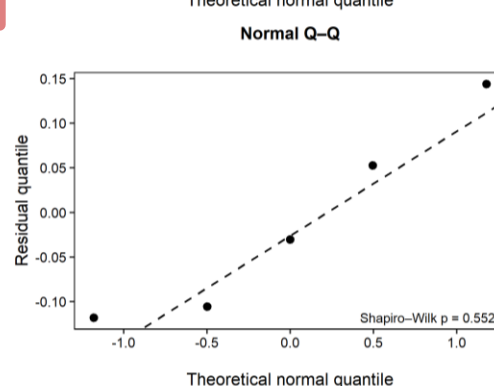
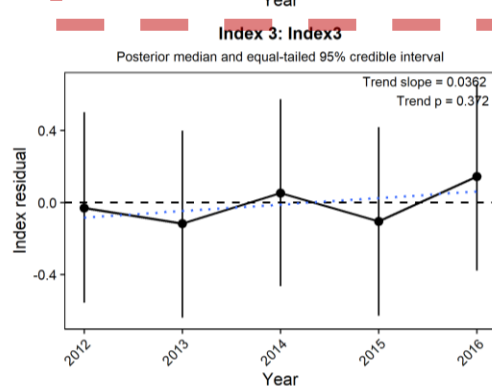
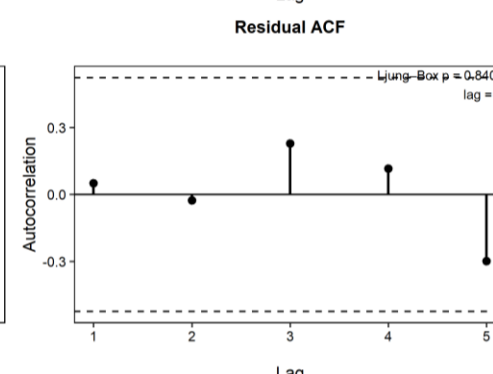
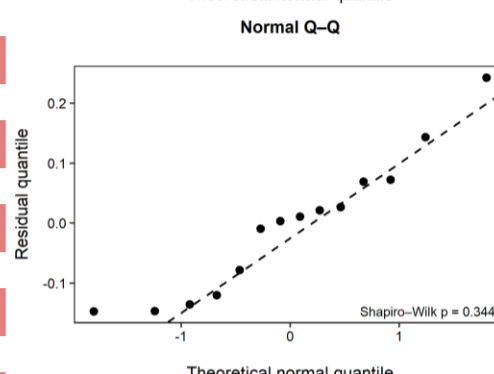
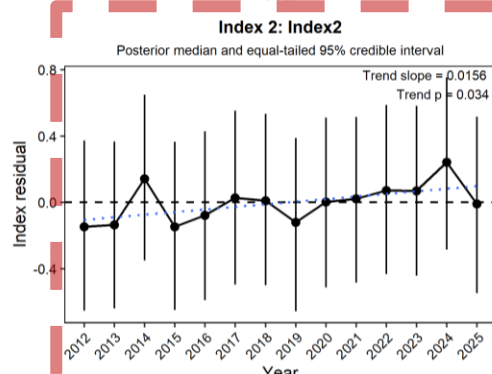
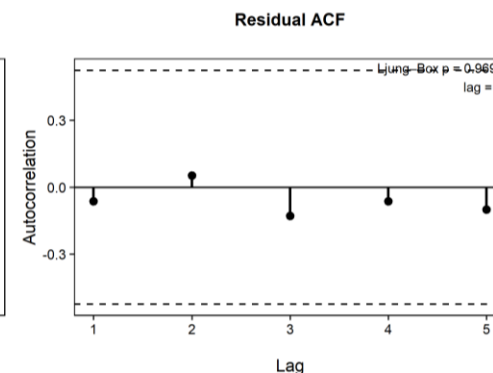
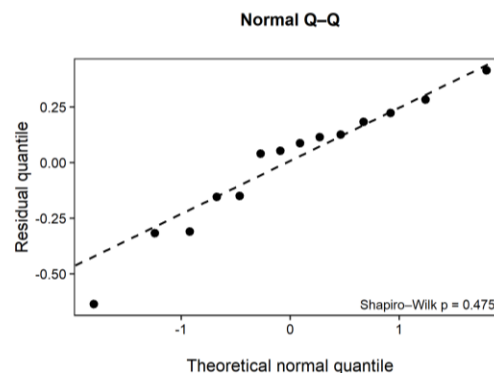
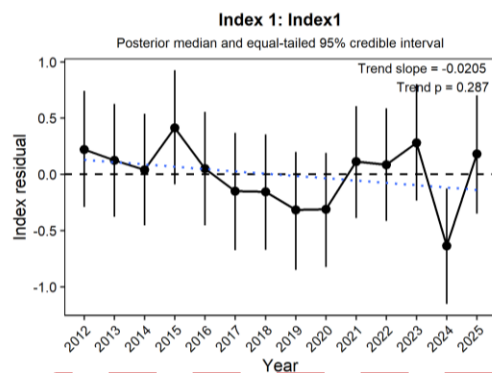
Model diagnostic

SE1



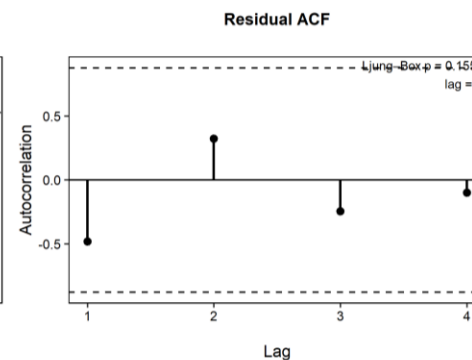
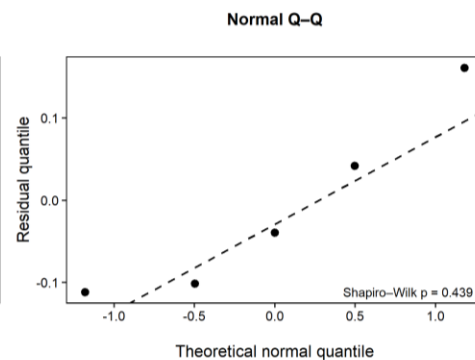
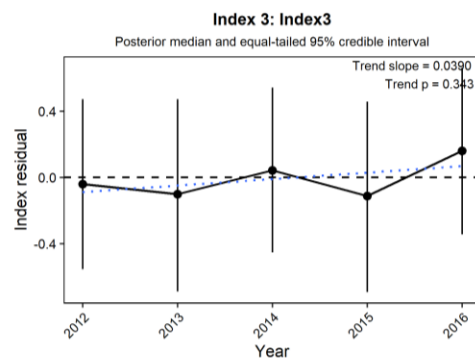
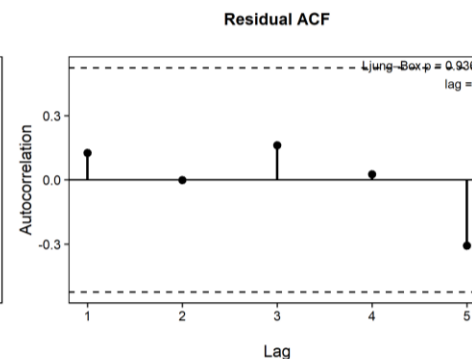
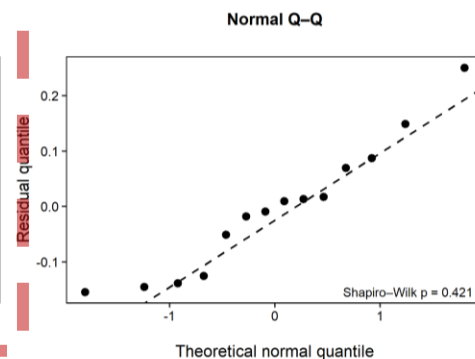
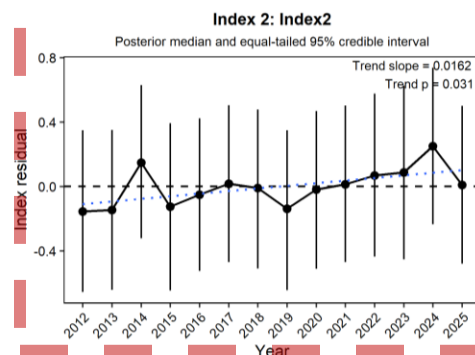
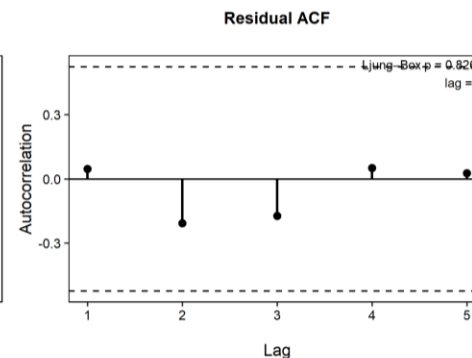
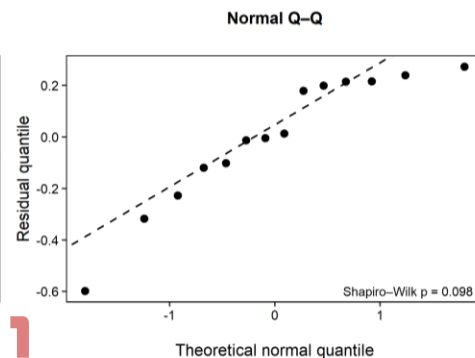
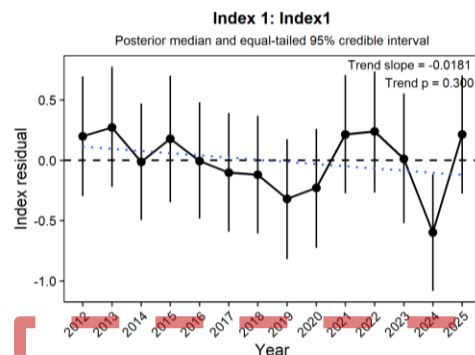
Model diagnostic

SE2



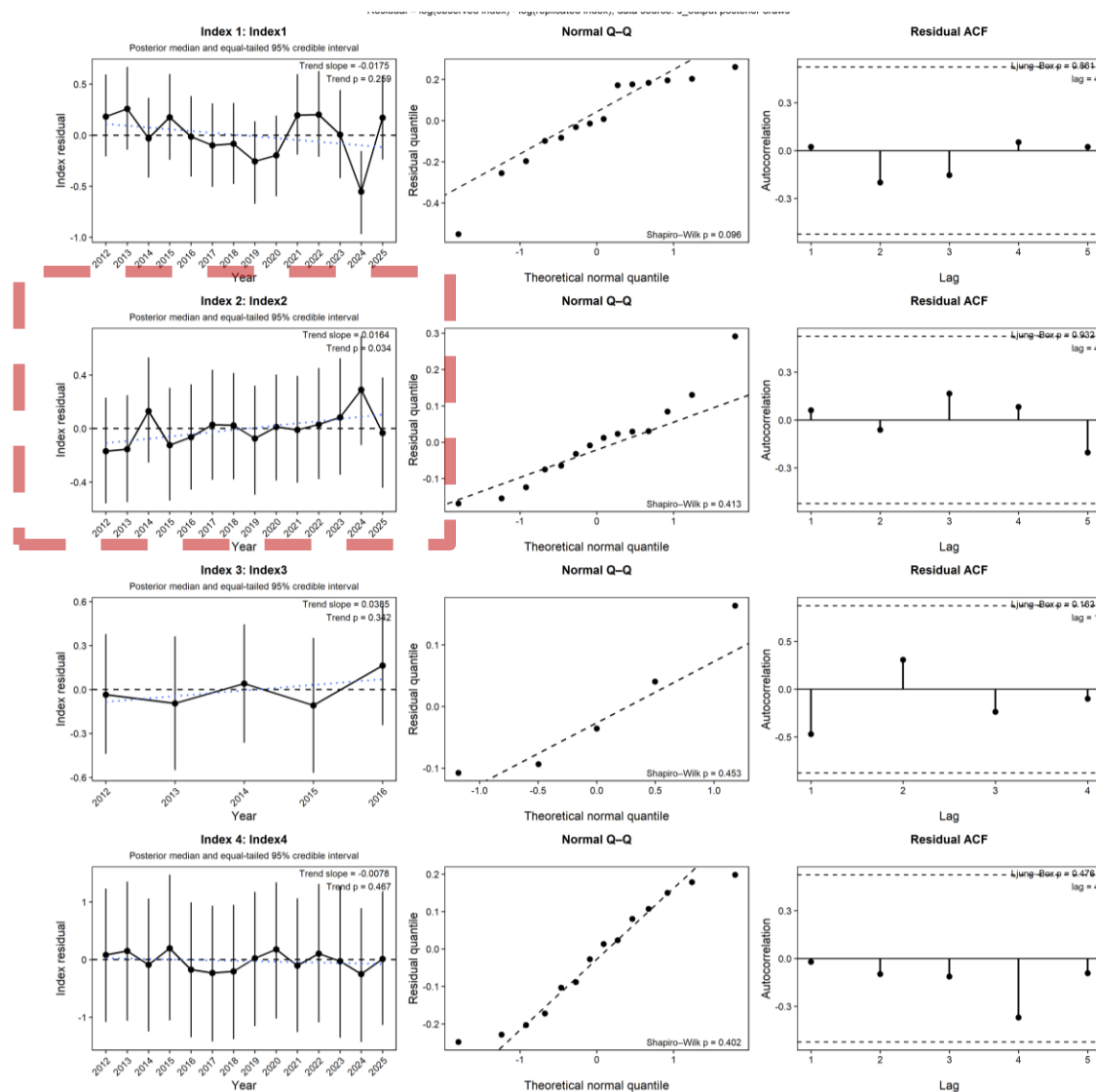
Model diagnostic

SE3



Model diagnostic

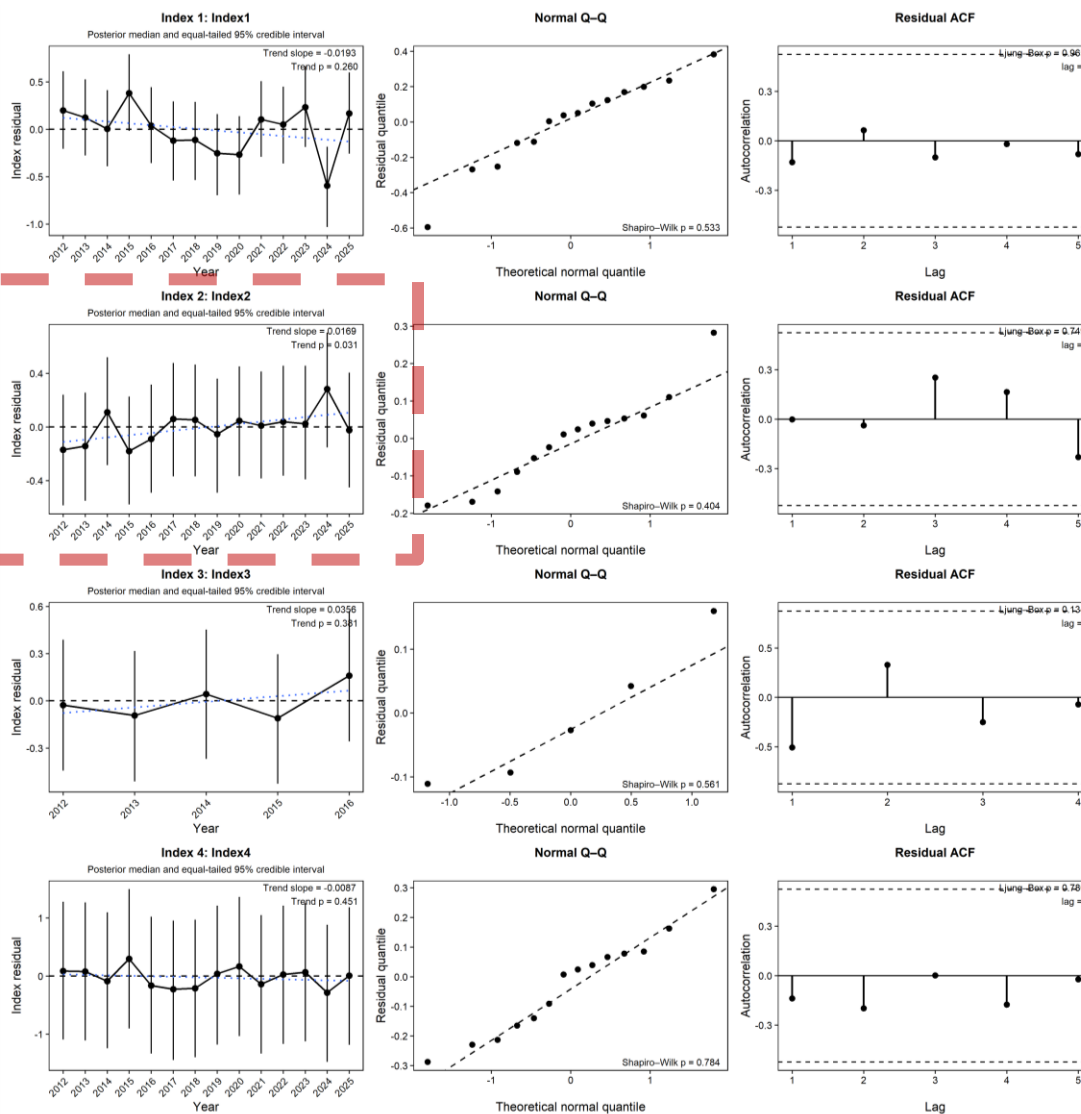
SE4





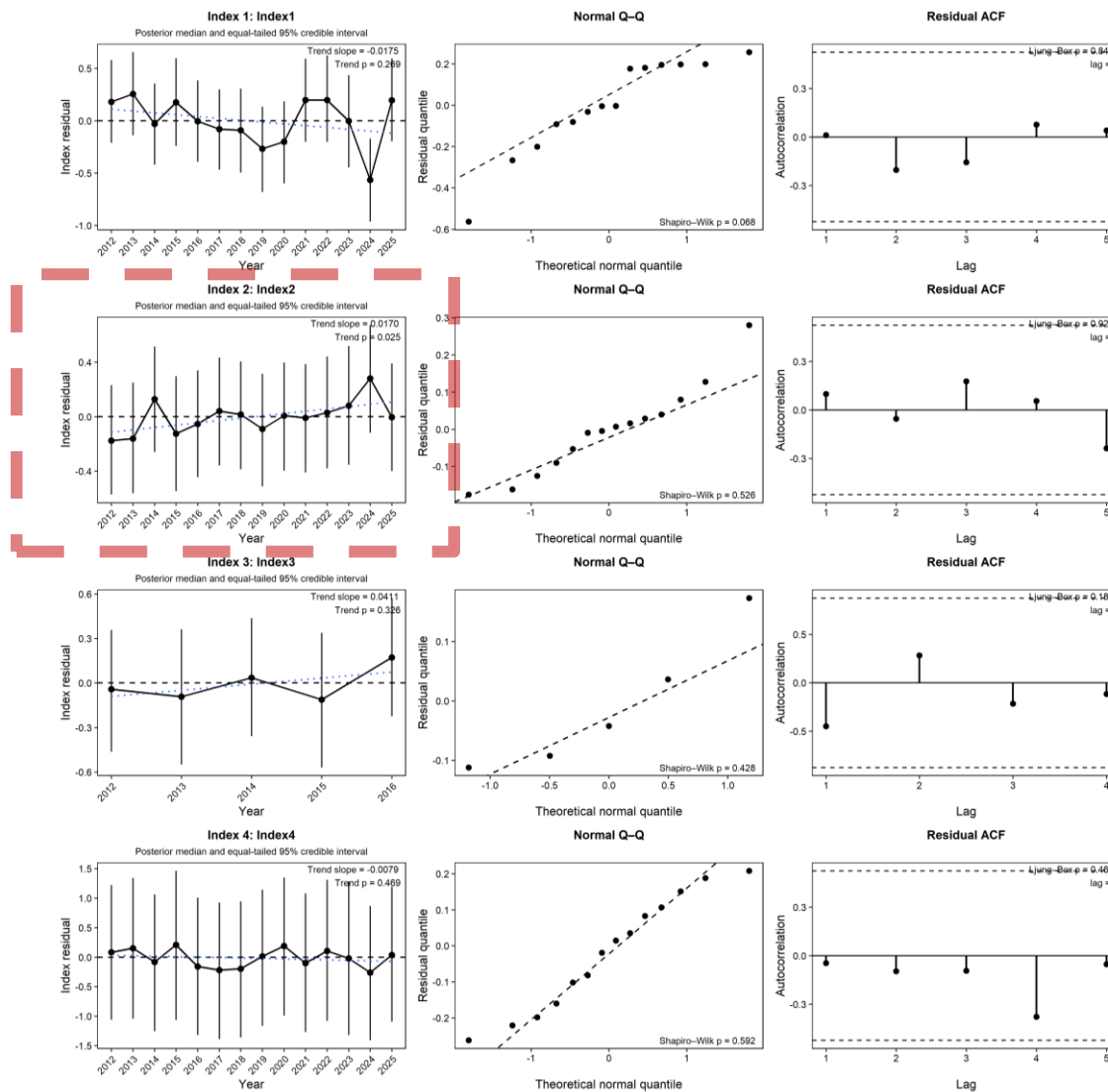
Model diagnostic

SE5



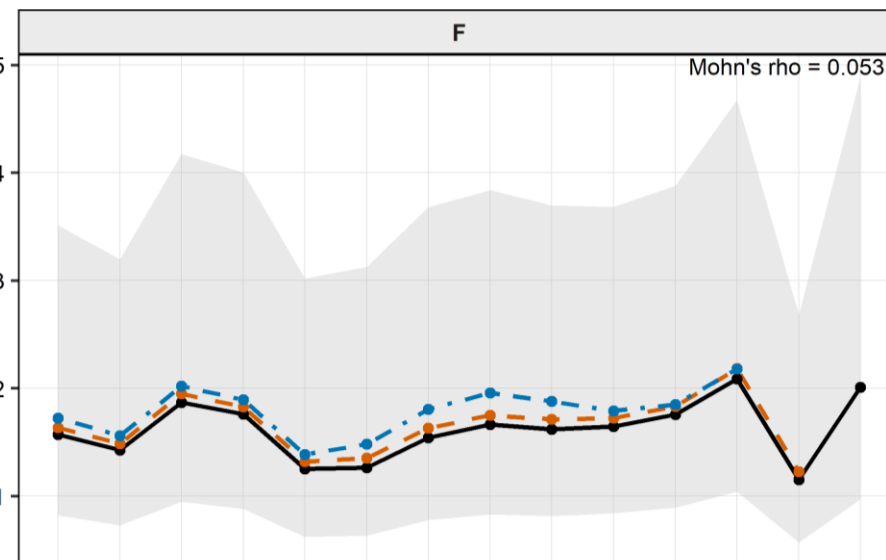
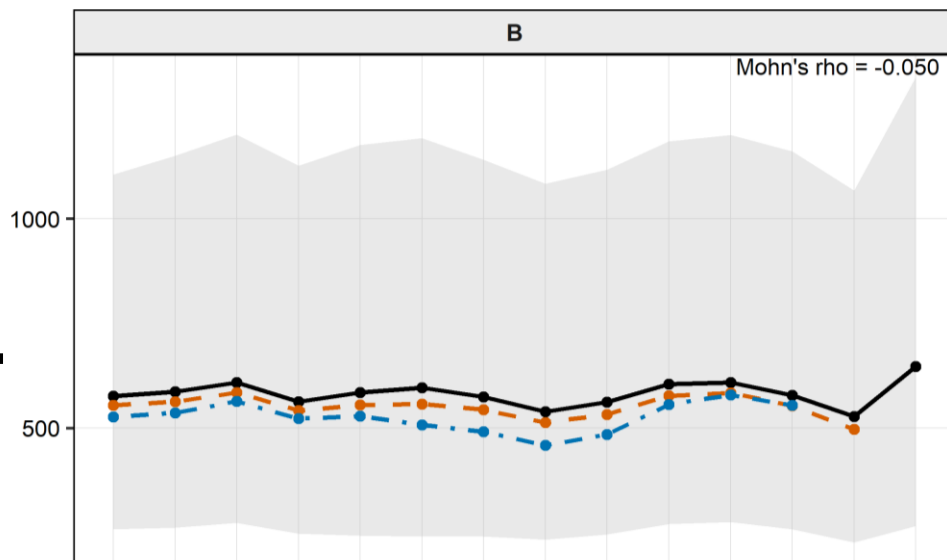
Model diagnostic

SE6

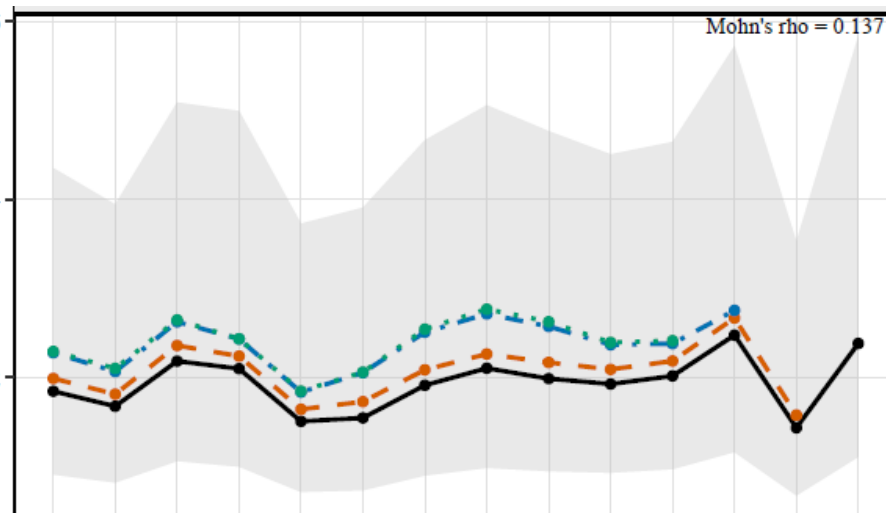
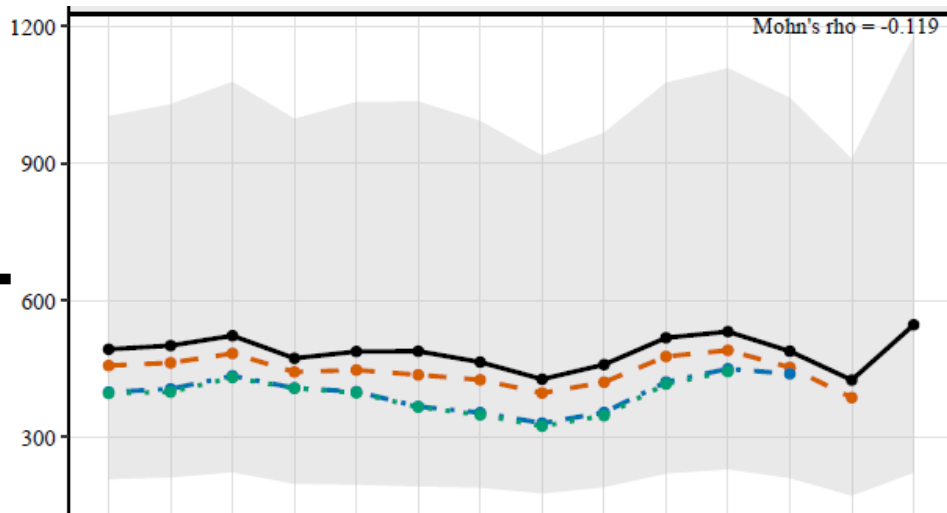


Model diagnostic

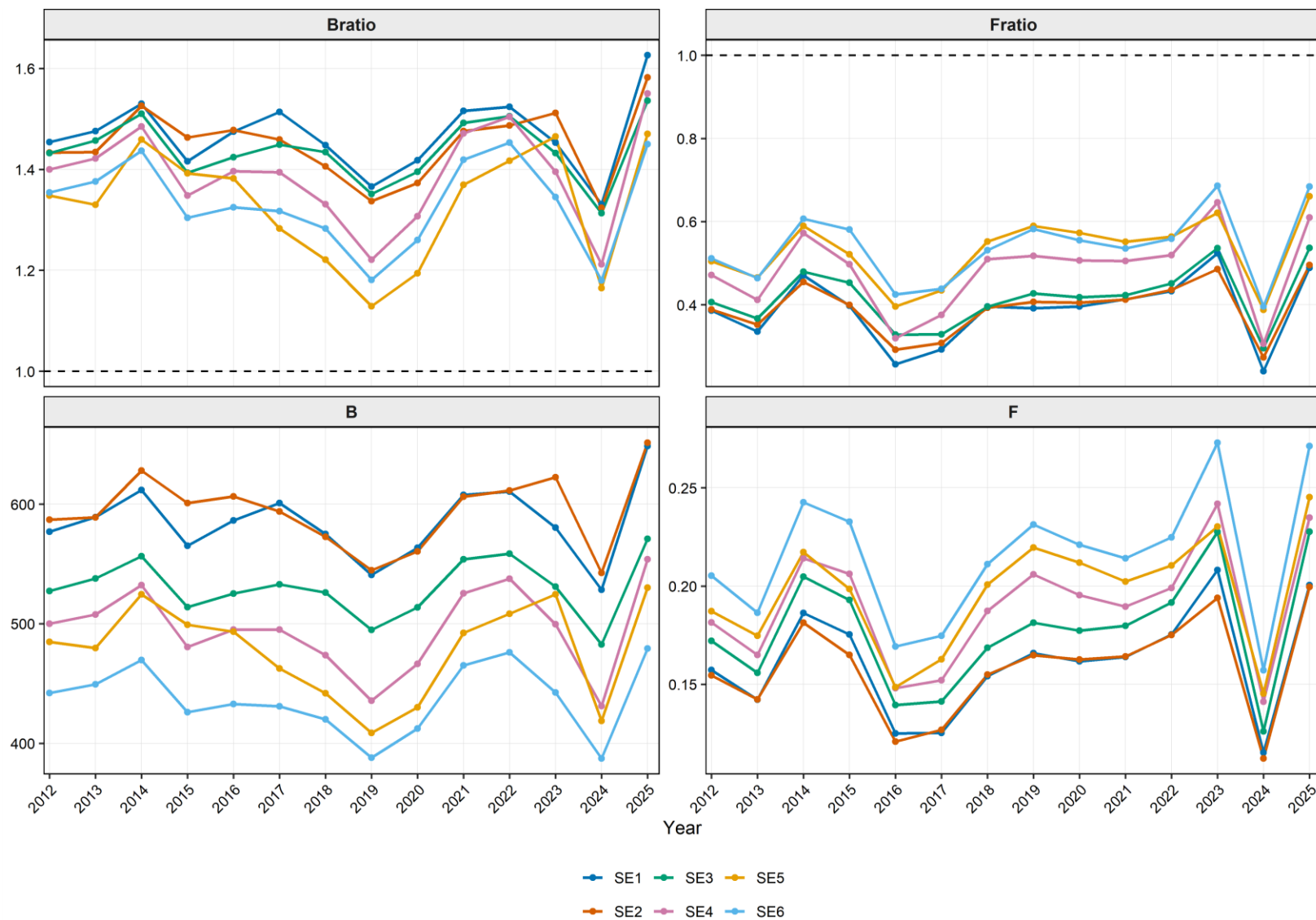
SE1



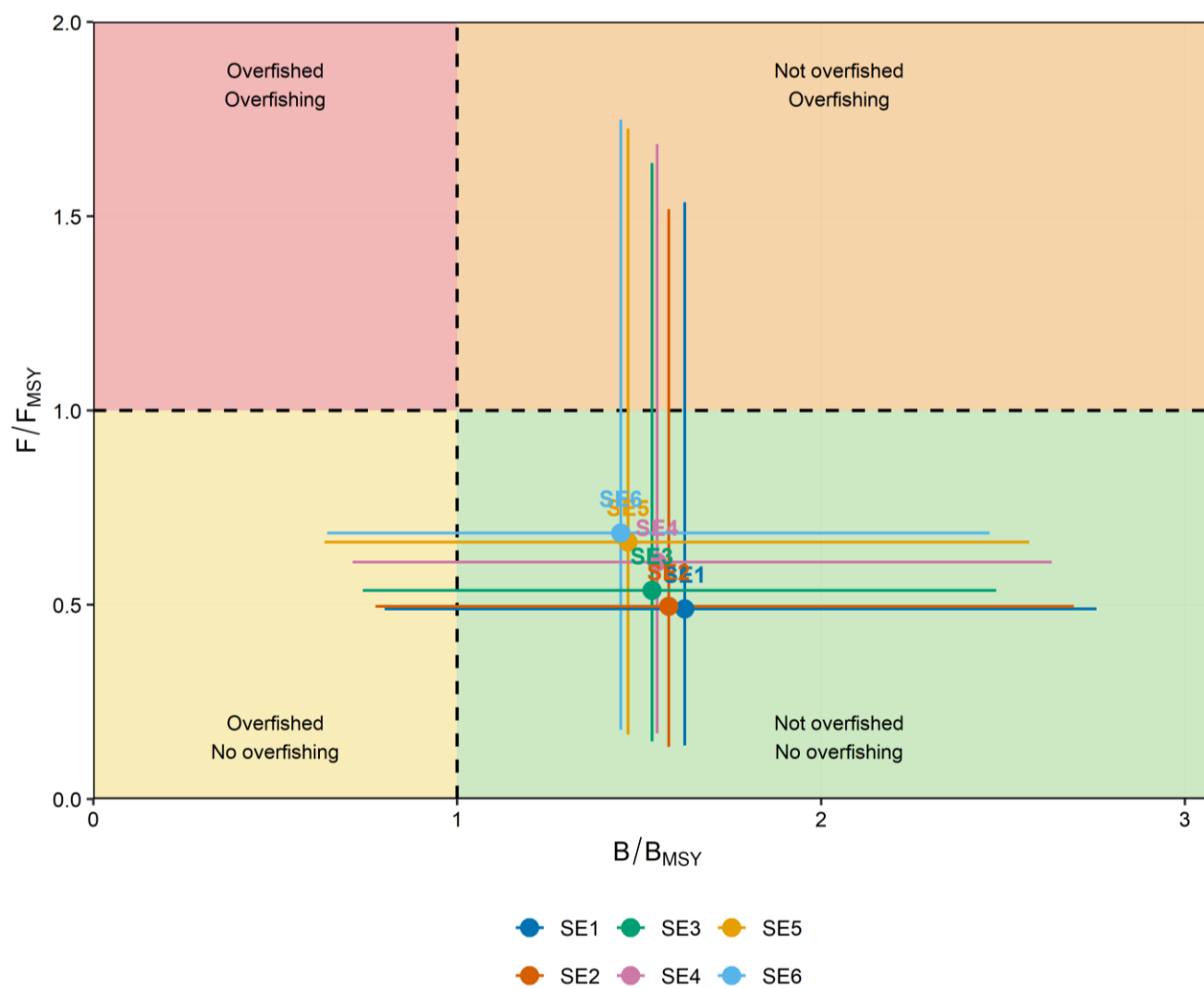
SE4



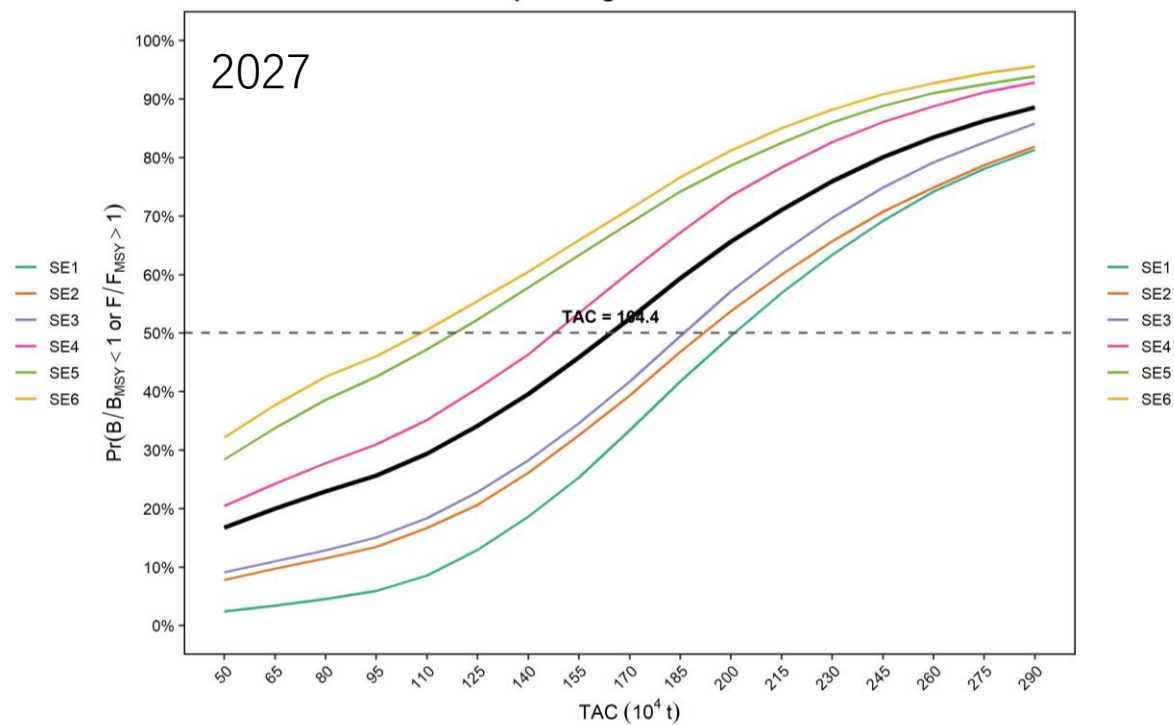
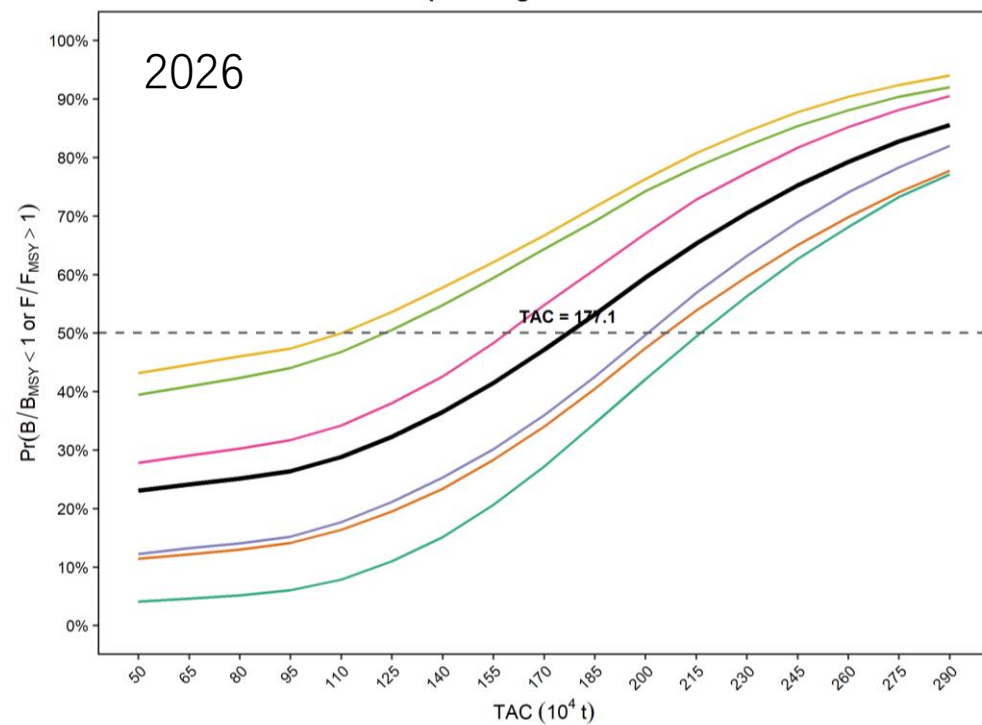
Model result



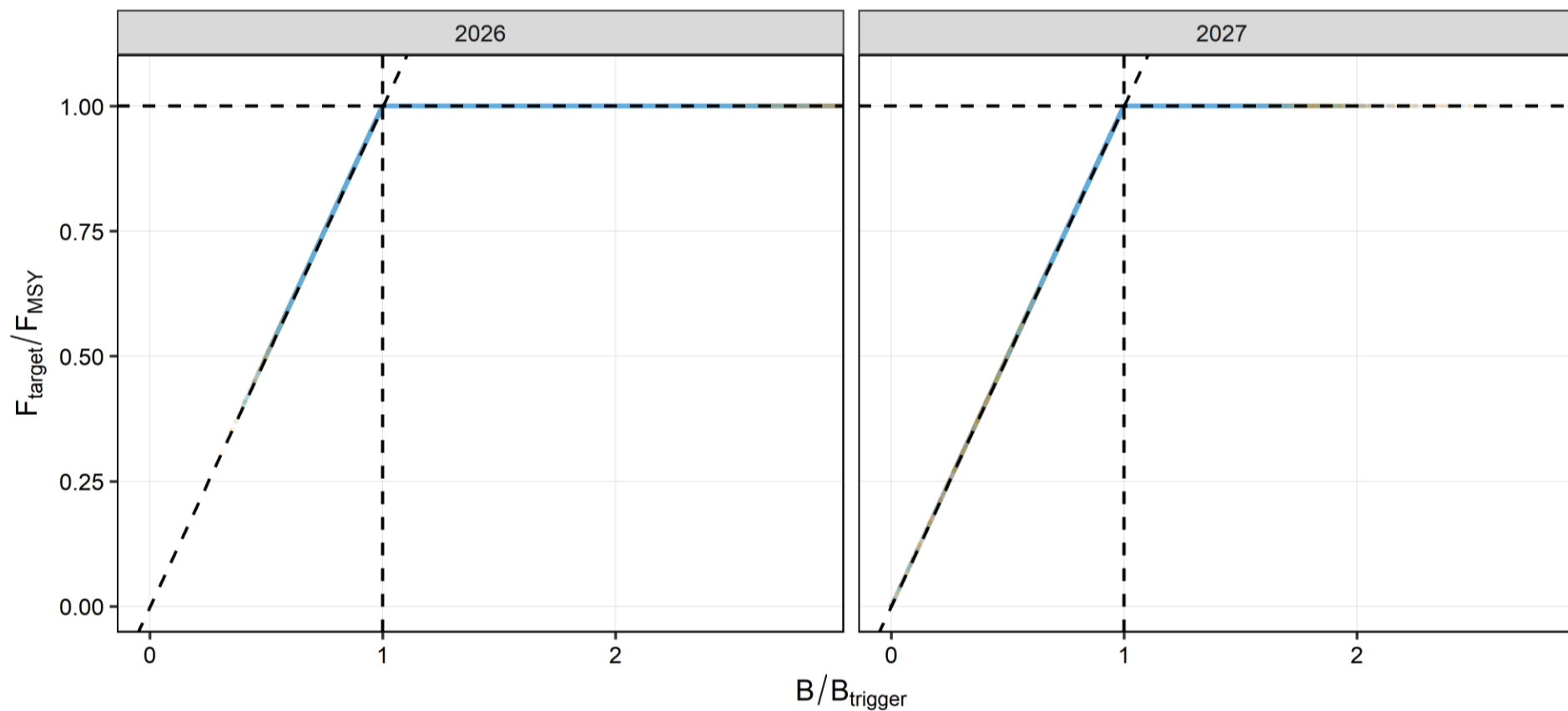
Model result



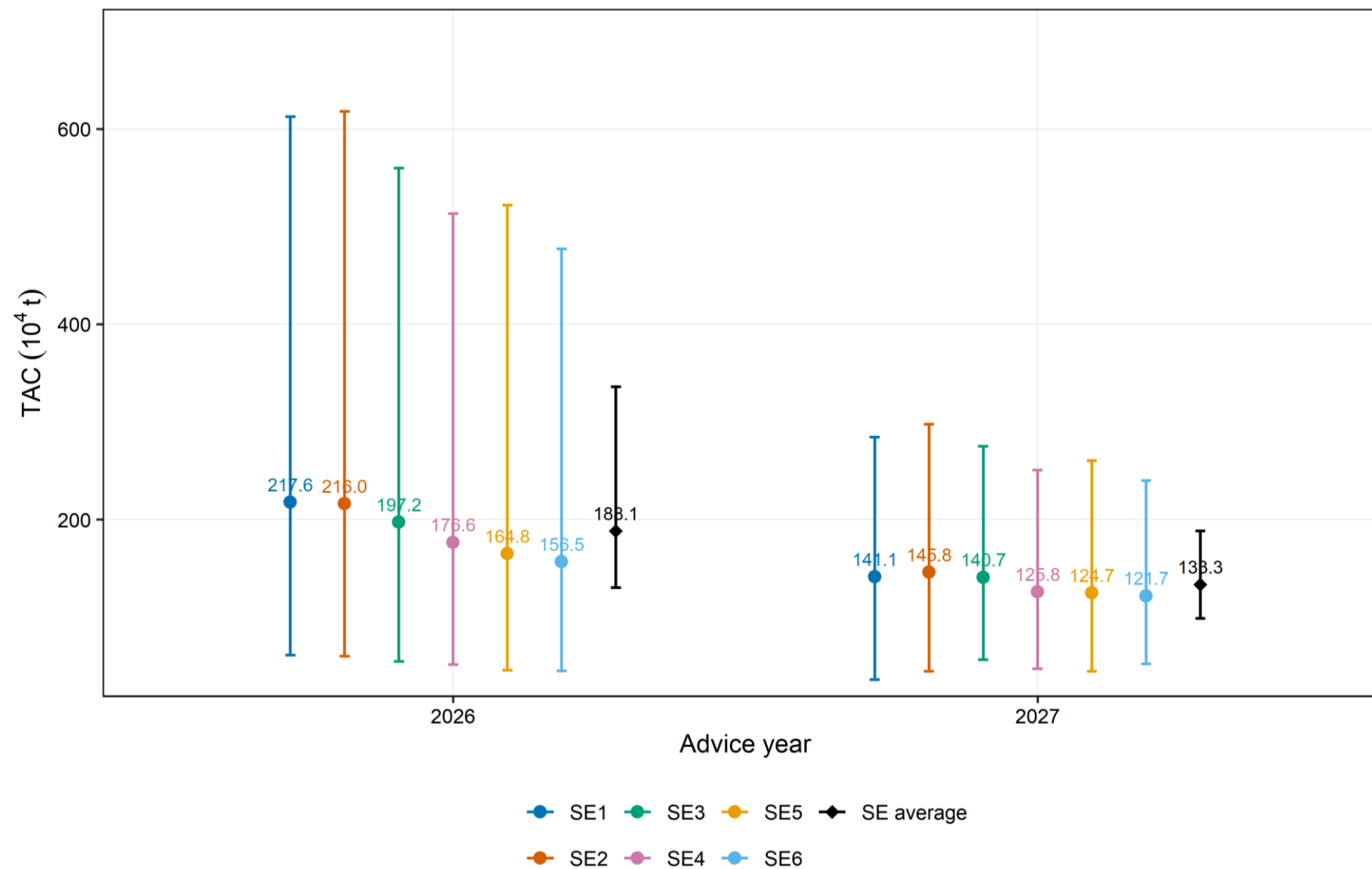
Projections



Model description



Model description





Conclusion

- ◆ The stock is in healthy status in 2025
- ◆ A TAC of 1881 thousand tons in 2026 and 1383 thousand tons in 2027 following the hockey method
- ◆ A TAC of 1771 thousand tons in 2026 and 1644 thousand tons in 2027 following the average method



THANK YOU FOR WATCHING

Mingfeng Jiang Gang Li Wei Yu Xinjun Chen
Shanghai Ocean University

07/29/2026

> GOODBYE <